

The Distorted Time Framework

Volume III — *The Definite Clock*

The Born rule, measurement, and entanglement from a clock that stays definite

William de Dufour

Independent Researcher, Austin, TX · billd.dtf@gmail.com

July 2026 — Version 8

DOI: [10.5281/zenodo.21659853](https://doi.org/10.5281/zenodo.21659853)

Abstract

Quantum mechanics postulates the Born rule and has never derived it; the closest route — Nelson’s stochastic mechanics — has stalled on Wallstrom’s single-valuedness obstruction and on the unexplained origin of its noise. Taking time’s local rate as primitive (Volume I), this volume reads a quantum’s phase as the clock’s reading and *reduces* the Born rule to the relaxation equilibrium of the conserved location density under the native guiding flow. Wallstrom is reduced to a single *declared* posit — that Row 0’s beable is a single-valued amplitude assignment — for the simply-connected sector, the nodal case left open. Nelson’s missing noise is named: the internal flexing (a zitterbewegung) of bound energy at the Compton scale, fixing the diffusion coefficient to $\hbar/2m$; its conservative (time-symmetric) character — hence Schrödinger rather than a heat equation — follows from the flexing being a *bound* mode, which cannot dissipate, on a timeless substrate. Entanglement is a shared present on the proper now, with CHSH = $2\sqrt{2}$ inherited, not derived. Two honest limits are kept in view: the guiding wave lives on configuration space ($O(N^2)$ content against a slice’s $O(N)$), so the local-beable reading holds only for one particle; and the amplitude sector is non-relativistic, its relativistic completion (toward Dirac) an open program. The claim throughout is reduction and identification, not a derivation of quantum mechanics.

Keywords: Born rule; Nelson stochastic mechanics; Wallstrom obstruction; pilot-wave theory; zitterbewegung; entanglement; measurement.

The three commitments (from the Core, Volume I). This volume builds on three, argued there and used here without re-argument (full recap: Appendix A):

- **1. Time is primitive** — its local rate u is the one physical thing, and a quantum’s phase is its reading, $\Phi = -\omega_C \int u dt$ ($\omega_C = mc^2/\hbar$): Row 0 (timeless, $\hat{H}\Psi = 0$) → Row 1 (the definite clock) → Row 2 (space); “becoming definite” is the Row-0→1 step.
- **2. Space is time’s radiation** — and the moving clock’s momentum is $\mathbf{p} = \nabla S = m\mathbf{v}$: inertia *is* the rate ω_C , the pilot-wave guidance the same space-face gradient ∇S ($\hbar$ -coupled, present even at $u \equiv 1$); the G -coupled well the rate digs is gravity, a separate object.
- **3. There is a proper now** — the preferred slice a non-propagating rate fixes, real but empirically silent; here it is the shared present that carries entanglement.

Here the rate’s reading is the quantum phase, and its residual internal flexing is the quantum noise.

Quantum mechanics postulates the Born rule [1] and has never derived it. The route that came closest — Nelson’s stochastic mechanics [3] — has been held up thirty years by one obstruction (Wallstrom’s [10]: nothing forces the phase to be single-valued, so the quantised spectrum is lost). This volume **reduces Born to a relaxation equilibrium on three physical premises** — and *reduces* the Wallstrom obstruction to a single ontological posit rather than an imposed quantisation

rule: take the fundamental object to be Row 0's amplitude assignment ($\hat{H}\psi_0 = 0$), a single-valued function on configuration space, and integer circulation then follows by topology alone, with nothing added by hand. That posit is *declared*, not derived — and it is a genuine restriction, since physically multivalued states exist (spinors, anyons) — so the reduction is made for the simply-connected sector and the nodal / multiply-connected case stays open. It keeps de Broglie–Bohm's [2] commitment that the particle is real and guided. On the wave equation itself the claim is bounded and stated once: DTF derives the *phase* half of the Schrödinger equation — the space-face of its own interval — and *reduces* the amplitude half to a single posited scale, $\hbar$, the clock's jitter; it does not claim to derive the equation, and the closing box totals this at its true size.

It does *not* make the guiding object local. Counting settles this rather than leaving it open: a general two-particle amplitude needs functional content growing like N^2 on N cells of space, while everything DTF puts on a slice — the clock rate and the extent structure — grows like N . No bounded amount of internal structure closes that gap, and enlarging it until it does is configuration space with an index renamed. So the guiding wave lives on configuration space here as in every Bohmian theory; the local-beable reading holds for a single quantum and does not extend [DTF_sharednow_counting.py].

What is distinctive survives that, and is the volume's actual claim. Relativistic Bohmian mechanics needs a preferred foliation and must postulate one. DTF's is not postulated: it falls out of $\pi_u \equiv 0$, and it is the same slice on which gravity already solves its field — introduced in Volume I for reasons that have nothing to do with Bell. The economy is in where the foliation comes from, not in making the wave local.

Chain 1 *The wave is a phase field, not a vanishing*

The particle stays definite and stays somewhere. What “spreads” is not the particle but the phase field that guides it — a gradient in space, not a wake in the rate of time.

$$\psi = R e^{iS/\hbar}, \quad \Phi = -\omega_C \tau \text{ (the clock reading)}$$

[known: polar form; phase as elapsed proper time]

Known

$$S = \hbar \Phi = -mc^2 \int d\tau : \quad \mathbf{p} = \nabla S = \gamma m \mathbf{v} \text{ (space-face)}, \quad E = -\partial_t S = \gamma mc^2 \text{ (time-face)}$$

[known: Hamilton-Jacobi: the one phase has two faces ($\int u dt$ at rest)]

Known

a clock carries two independent things: a *rate* (its mass ω_C , which moves it and bends time around it) and a *state* (a spin). Entanglement touches only the state.

Derivation

singlet: spin purity 0.5, each rest energy sharp, spread 0.0000

`DTF_two_clocks_resolved.py` → rate untouched

DTF result — verified

$u \equiv 1 \Rightarrow \nabla S = \gamma m \mathbf{v} \neq 0$ (guidance is the space face)

`DTF_guidance_is_spaceface.py` → no G , no u -wake

DTF result — verified

the “wave” = the phase gradient ∇S , $\nabla S \neq 0$ even at $u \equiv 1$

DTF result

The particle never de-materialises. What guides it is the space-face phase gradient ∇S — which survives even with the clock rate perfectly uniform ($u \equiv 1$), so it is no wake in u . The rate itself (the mass ω_C) is the separate thing entanglement leaves razor-sharp.

Takeaway

One object, seen here as: inertia & the pilot wave

Both read the one phase $\Phi = -\omega_C \tau$, and *neither carries G* . Inertia *is* the rate ω_C — the mass $m = \hbar \omega_C / c^2$, linear in m , definitional: the clock's ticking, not a mechanism. Guidance is the *space*-face gradient ∇S — $\hbar$ -coupled, and it *survives at $u \equiv 1$* (a flat clock field). The two are tied because the momentum *is* that gradient, $\mathbf{p} = \nabla S = m\mathbf{v}$: one clock, read as its rate and as the gradient it lays down. What is *neither* — the G -coupled well the rate digs in u , $\sim (m/m_P)^2$ — is gravity (Volume II). Met again through Volumes I and II.

Chain 2 *Amplitude is the same clock, read on the space axis*

The phase read along time is a period; along space it is an extent. One constant ties them.

$$E T_C = h \text{ (phase)}, \quad E \lambda_C = hc \text{ (amplitude)}, \quad \lambda_C = c T_C$$

[known: Planck/Compton, one on each axis (Volume I, Chain 5)]

Known

h and hc differ only by c

DTF result

The phase-anchor and amplitude-anchor are one constant, not two — “space is time’s radiation.” The pilot-wave tradition reads the time face, textbook Born the space face; the clock supplies both.

Takeaway

The most legible line in the framework: the double slit. With $\Phi = -\omega_C \tau$, bright fringes sit where the two routes’ accumulated *proper times* differ by a whole number of Compton periods — constructive interference is $\omega_C \Delta\tau = 2\pi n$, i.e. $\Delta\tau = n T_C$. For two paths reaching the same lab time this is *exactly* de Broglie’s $\Delta\ell = nh/p$ [DTF_doubleslit_compton_ticks.py, verified exact — not small-angle]. Label it honestly: the relativistic action phase rewritten in clock language, a *consistency check*, not new physics. Its worth is that for *one* particle configuration space *is* ordinary 3-space, so the single-particle interferometer is where DTF’s local-beable reading holds with **nothing owed** — its best case, not its hardest. (Two particles is Chain 4, where the debt begins.)

Chain 3 *The Born rule as the flow's equilibrium*

$|\psi|^2$ is not a probability postulate — it is the density of where the definite particle is. So compute where it settles.

$$\rho = |\psi|^2, \quad \partial_t \rho + \nabla \cdot (\rho \mathbf{v}) = 0$$

[known: polar form + local conservation]

Known

[DTF input:] the particle flows along its guiding phase gradient, $\mathbf{v} = \nabla S/m$ — the same ∇S that is its momentum, not a new law

DTF input

conservation makes $\rho = |\psi|^2$ the *stationary* distribution of the flow (equivariance); the settling toward it is a coarse-grained H -theorem. The remaining ingredient is single-valuedness — Wallstrom's obstruction, that the hydrodynamic pair (ρ, S) does not force $\oint \nabla S \cdot d\ell \in 2\pi\hbar\mathbb{Z}$. Here it is not imposed as a separate quantisation rule but *reduced to one ontological posit*: take DTF's fundamental object to be Row 0's amplitude assignment ψ_0 ($\hat{H}\psi_0 = 0$), a single-valued function on configuration space, rather than the Madelung pair. Then a non-integer-winding mode assigns two values to one configuration — it is not such a function — and integer circulation follows by topology, with nothing added at Row 1 (conditioning on the clock cannot manufacture a winding ψ_0 never had). The posit does the work a by-hand quantisation rule used to do; it is *declared*, not derived, and it is a genuine restriction — physically multivalued states exist (spinors, anyons) — so the reduction is made for the simply-connected sector, and the nodal / multiply-connected case, where multivalued states can be physical, stays open.

Derivation

start at $|\psi|^2 \Rightarrow$ stays $|\psi|^2$ (equivariance)

`DTF_born_relaxation_2d.py` $\rightarrow H \approx 0.01$

DTF result — verified

start off-equilibrium $\Rightarrow$ relaxes to $|\psi|^2$

`DTF_born_relaxation_2d_mp.py` $\rightarrow H: 0.78 \rightarrow 0.04$

DTF result — verified

single-valued $\psi_0 \Rightarrow \oint \nabla S \cdot d\ell = 2\pi n\hbar$ (simply-connected sector)

`DTF_wallstrom_from_row0.py` $\rightarrow$ reduced to one posit

DTF result — verified

$|\psi|^2$ = the equilibrium the conserved location-density settles onto

DTF result

Born is reduced to three premises (closing ledger), not postulated outright: it is the fixed point of the guiding flow, reached by relaxation, with the spectrum quantised because single-valuedness is reduced to a single Row-0 posit rather than imposed by hand. Which outcome a single run yields — the hard half of measurement — stays open. And the direction matters: this chain assumes ψ obeys the Schrödinger equation and shows $\rho \rightarrow |\psi|^2$, so it reduces Born downstream of that equation — it is not a route to deriving the equation (the closing box totals what is and is not derived).

Takeaway

Chain 4 Entanglement is a shared present, not a signal

Bind quanta and you get one object with several addresses. The link is a ledger, not a channel.

singlet: part purity $\text{tr } \rho_A^2 = 0.5$

[known: joint definiteness without individual definiteness]

Known

the correlation is a Row-0 constraint (a conserved total, no location), balanced on the *proper now* — the shared present. Separating the parts stretches nothing; measuring one reads an entry. Nothing crosses the gap.

Derivation

GHZ: measure one $\Rightarrow$ other two definite *or* a fresh pair; marginals unchanged

`DTF_tripartite_ledger.py` $\rightarrow$ no-signalling

DTF result — verified

DTF's *local* wake in ordinary space gives CHSH = 2 — the local-beable bound (Bell 1964 [9]; Maudlin [16]; Norsen [17] — a theorem, not a misreading). To exceed it the correlation must ride the *shared present*: a joint state on one preferred slice. Relaxing that joint state to its Born measure reproduces CHSH = $2\sqrt{2}$ [`DTF_bell_from_joint_relaxation.py`] — but this is standard foliation-based relativistic Bohmian mechanics (Dürr–Goldstein–Norsen–Zanghì [18]; Tumulka [19]), and the $2\sqrt{2}$ is *inherited* from the entangled state, not generated by the relaxation (which only drives ρ toward a given $|\psi|^2$).

Derivation

the shared now is the *resource* for Bell violation — and DTF gets it for free

DTF result

The claim is not a new derivation of $2\sqrt{2}$ but an ontological economy over the foliation-based relativistic Bohmian programme: those theories postulate the preferred foliation externally, whereas DTF's — the proper now — falls out of $\pi_u \equiv 0$ (Volume I). That is the loud, defensible claim, and the only one made here. Whether the shared-now joint object is distinguishable from configuration space under another name is settled, against the local reading: it is not. A general two-particle amplitude needs $O(N^2)$ functional content and a slice carries $O(N)$, a gap no bounded internal structure closes. So the joint object is configuration space, and this chain claims no locality it cannot have.

Takeaway

One object, seen here as: the proper now — a shared present

The same preferred slice on which gravity solves its field (Volume I, Chain 7) is where an entangled system's joint state — one object on configuration space, not two local clocks — is referred to a single present. One slice, two jobs; and because c is universal it stays undetectable — no signal, no broken relativity.

The Schrödinger equation, reassembled — and where it stops. In polar form $\psi = R e^{iS/\hbar}$ the Schrödinger equation splits *exactly* (Madelung [8]) into a phase law and an amplitude law:

$$\underbrace{\partial_t S + \frac{(\nabla S)^2}{2m} + V + Q = 0}_{\text{phase: Hamilton-Jacobi}}, \quad \underbrace{\partial_t R^2 + \nabla \cdot \left(R^2 \frac{\nabla S}{m} \right) = 0}_{\text{amplitude: continuity}}, \quad Q = -\frac{\hbar^2}{2m} \frac{\nabla^2 R}{R}.$$

DTF_madelung_split.py → the split, exhibited — phase half native, amplitude half imported

The phase law is DTF-native, and reassembles from the chains above. Its pieces are already deposited: the accumulated clock reading $S = -mc^2 \int u dt$ with its two faces $\mathbf{p} = \nabla S$ and $E = -\partial_t S$ (Chain 1), and V from the clock's coupling to matter (Volumes I–II). Collected, they are the *classical* Hamilton–Jacobi equation — missing exactly Q . **The amplitude law is imported.** The continuity equation for R^2 is Chain 3's premise, and with it Q — which carries *all* the quantum content and is built entirely from R . So the quantum lives wholly in the half DTF does not derive — and here the framework says something definite about it. **This is Nelson's stochastic mechanics [3], with the one thing Nelson left unexplained now named.** Nelson recovers exactly this amplitude law from an underlying conservative diffusion of coefficient $\nu = \hbar/2m$, but must *posit* the diffusion; its physical origin was never supplied. DTF supplies it — and *not* at the Row-0→Row-1 interface, which carries only the conservation law and single-valuedness (below). The noise is intrinsic to Row 1: a particle *is* bound energy, and it flexes as it oscillates inside itself at the Compton clock frequency $\omega_C = mc^2/\hbar$ — the trembling internal motion of a *zitterbewegung* [4, 5]. That flexing is unresolvable at the position level, and, carried into the phase $\Phi = -\omega_C \int u dt$, *is* the sub-quantum noise. So the quantum potential Q is not imported machinery but the shadow of the internal flexing, and the classical limit is the particle whose internal motion is ignorable ($Q \rightarrow 0$). And because that sub-tick

flexing can *never* enter a position-level deterministic formula, it is at one stroke the origin of quantum *indeterminism* — the piece Nelson takes as brute noise.

The scale, and the factor of 2. The flexing amplitude is the Compton cell $\lambda_C = \hbar/mc$, one oscillation per tick $T_C = \hbar/mc^2$; an unbiased random walk of that step has the *standard* diffusion coefficient

$$\nu = \frac{\lambda_C^2}{2T_C} = \frac{\hbar}{2m},$$

which is Nelson’s ν exactly — though the factor of 2 carries the one-dimensional random-walk convention $\langle x^2 \rangle = 2\nu t$, and is not itself fixed by the flexing dynamics. So $\hbar$ is *reparented*: not posited afresh as “the quantum of action” but read as the scale of the particle’s own internal flexing. Writing the step as $\alpha\lambda_C$ ($\nu = \alpha^2\hbar/2m$) and requiring the osmotic potential it builds, $-2m\nu^2\nabla^2 R/R$, to be the Madelung $Q = -\frac{\hbar^2}{2m}\nabla^2 R/R$, returns $\alpha = 1$: a *consistency check* that the two roles of $\hbar$ agree, not an independent derivation of the coefficient. What remains open — and is listed as owed below — is whether the flexing dynamics *forces* $\hbar/2m$ rather than merely admitting it. `DTF_nelson_from_flexing.py` $\rightarrow \nu = \hbar/2m$ (1D convention); `osmotic-match` returns $\alpha=1$ — consistency, not derivation

Conservative, not dissipative — and that is the whole of it. Nelson’s construction yields Schrödinger rather than a heat equation only if the diffusion is *conservative* (time-symmetric), his most-criticised assumption. Two DTF facts point the same way, and a free-particle check confirms the consequence: the internal flexing is a *bound* oscillation — rest mass neither radiates nor decays — and a bound, finite motion cannot dissipate (dissipation needs a thermal continuum), so it is reversible; and Row 0 is timeless, with no substrate arrow. For the free particle the time-symmetric (Nelson) acceleration vanishes *exactly* when the osmotic coefficient equals $\hbar/2m$, reproducing quantum spreading, whereas the forward-only (arrow-ful) rule gives the heat equation [`DTF_nelson_from_flexing.py`]. Whether the flexing dynamics realises this *exactly* is the sector’s one open item — stated as its single posit in the box below.

The interface, separately, carries conservation and single-valuedness. Distinct from the Row-1 flexing, the Row-0 $\rightarrow$ Row-1 conditioning is what supplies the continuity law (Chain 3) and preserves single-valuedness of ψ (Chain 3’s Wallstrom posit — it manufactures no winding). Noise on the amplitude; conservation and topology at the interface: two different jobs, not to be run together. **At true size:** DTF does not *derive* the Schrödinger equation. It supplies the phase half structurally and *reduces* the amplitude half to a single physical postulate — the conservative internal flexing of Row-1 bound energy — with the coefficient now native ($\hbar/2m$, the random-walk normalisation) and the flexing’s time-symmetry now shown to follow from bound-energy and a timeless Row 0 (harmonic model; only full nonlinear rigor remains — see the posit box). And *reducing Born is not deriving Schrödinger*: Chain 3 assumes ψ obeys the equation and shows $\rho \rightarrow |\psi|^2$ — downstream of it. What is new is an *identification*: Nelson’s unexplained noise, and quantum indeterminism with it, is the internal flexing of bound energy — the always-on, $\hbar$ -scale form of the same clock indefiniteness whose growth to an $O(1)$ crossing (there gravitationally driven) is measurement and the black-hole core. The trilogy marks the wall exactly here.

The one quantum posit — weighted with the singularity’s. Volume II’s black hole rests on a single ontological posit: Row-1 time *de-emerges* where the clock loses definiteness ($\delta u/u \sim 1$), and the finite core, the Planck remnant, and the absent echoes follow. The quantum

sector rests on one of the same weight. Given what the trilogy has already fixed — the density $\rho = |\psi|^2$, the phase S as the clock reading, and the flexing scale $\nu = \hbar/2m$ — the variational principle for the amplitude law (Madelung; Guerra–Morato [6]) has *one* free choice left: the sign of its osmotic term,

$\varepsilon = +1 \Rightarrow$ real-time, unitary: Schrödinger; $\varepsilon = -1 \Rightarrow$ imaginary-time, dissipative: a heat equation.

`DTF_qm_posit_variational.py` $\rightarrow \varepsilon=+1$ Euler–Lagrange $\Leftrightarrow$ Schrödinger (verified); $\varepsilon=-1 \Rightarrow$ heat That sign *is* the time-symmetry of the internal flexing — a flexing with no preferred time direction takes the real-time branch. So the whole quantum sector reduces to one statement:

The internal flexing of bound energy is time-symmetric.

And $\varepsilon = +1$ is not an arbitrary choice: *within a harmonic model of the flexing* it follows from what DTF holds. Model the flexing as its *own* (internal, not environmental) Caldeira–Leggett [7] sector: a *bound*, discrete oscillator has a vanishing friction kernel ($\tilde{\gamma}(s \rightarrow 0) = 0$, reactive only) and returns the coarse energy in *full* (reversible), so the induced diffusion is time-symmetric; and Row 0, being timeless, imports no arrow. *One caveat, stated plainly:* the physical flexing is a *zitterbewegung*, which in Dirac theory is interference across the positive/negative-energy *continuum*, not a few discrete modes — so whether it retains that time-symmetric character is part of the relativistic completion listed as open below, not settled here. Within the harmonic model, a vanishing friction kernel *is* a time-symmetric influence action, which *is* $\varepsilon = +1$ [`DTF_flexing_time_symmetry.py`]. So the quantum sector’s character is a *consequence* of the bound-energy ontology and a timeless substrate — shown here for the harmonic flexing model. It sits at the other end of the same axis as the singularity: the singularity is the *ceiling* of clock indefiniteness ($\delta u/u \rightarrow 1$, time de-emerges), the flexing its always-on *floor* ($\hbar$ -scale, definite but never perfectly). One knob, two ends. **The honest remainder** is the relativistic one: whether the physical flexing — a continuum *zitterbewegung*, not the harmonic model’s discrete mode — retains the time-symmetric character; that is the relativistic-completion question below, not settled here.

At relativistic speed: an open program, and where DTF stands. The amplitude law above is non-relativistic (Schrödinger), while the phase is relativistic ($\Phi = -\omega_C \int d\tau$, with $\mathbf{p} = \gamma m \mathbf{v}$). Closing that gap is an open program — but an unusually favourable one here, for two structural reasons. *First*, relativistic stochastic mechanics classically fails for want of a preferred time in which the diffusion runs; DTF already has one — the proper now ($\pi_u \equiv 0$), the same slice that does gravity and entanglement — so the natural covariant statement is that the diffusion runs in *proper time*, $\nu = \hbar/2m$ per $d\tau$, exactly as the phase does. *Second*, the noise is a *zitterbewegung*, a Dirac-equation phenomenon; so the relativistic amplitude sector should complete not toward a patched Schrödinger but toward *Dirac* (Hestenes [5]), with Schrödinger as its $\gamma \rightarrow 1$ face. *The honest ceiling:* the flexing amplitude *is* the Compton length λ_C — precisely the scale at which single-particle quantum mechanics gives way to pair production and quantum field theory. So the single-particle construction is intrinsically bounded below the Compton scale, and the full completion is ultimately field-theoretic, not merely Dirac. None of this is claimed here; it is the sharpest forward direction, and DTF holds the two ingredients the direction needs — a preferred time, and a relativistic noise.

What is earned, borrowed, declared, and owed.

- **Earned.** The phase as a clock reading; the one phase Φ as the shared root of inertia (the rate), gravity (the well it sources), and guidance (its space-face gradient); the amplitude as the same clock on the space axis; entanglement as joint-without-individual definiteness on the proper now; **the Born rule**, as the relaxation equilibrium of the conserved location-density (verified), with quantised circulation *reduced* to a single Row-0 posit (a single-valued amplitude assignment) rather than imposed by hand — for the simply-connected sector.
- **Borrowed.** The Tsirelson bound $2\sqrt{2}$ [12]; the sub-quantum H -theorem (Valentini [11]) and the diffusion construction of the amplitude sector (Nelson [3]).
- **Retired — not borrowed.** The measurement-axiom scaffolding (Gleason [14], Busch [15]) is not inherited here but *dropped*: those theorems justify $|\psi|^2$ as the unique measure over measurement outcomes, and once $|\psi|^2$ is instead the equilibrium of a conserved location density, that is a question this account does not ask. Stated as its own commitment rather than a clause, because it is the stronger claim: not that the scaffolding is used differently, but that it is unnecessary.
- **Declared.** The substrate is unresolved to any observer built from it (typicality); correlations are overlaps (exact for the qubit); and the posit that carries the Wallstrom reduction — that Row 0's beable is a *single-valued* amplitude assignment on configuration space. This is what replaces a by-hand quantisation rule; it is a genuine restriction (spinors and anyons are physically multivalued), so it is declared for the simply-connected sector, not derived in general.
- **Owed.** Coarse-graining (typicality); and *which* single outcome a run yields (selection), which every no-collapse ontology owes. *The amplitude sector's dynamics*: the phase half of the Schrödinger equation is DTF-native (the reassembly above), but the amplitude half is imported. For its origin DTF offers a *candidate*, not a derivation — the clock's cell-to-cell resynchronisation, whose fluctuation is a natural source for the diffusion the amplitude law needs — at a scale $\hbar$ posited as every formulation of quantum mechanics posits it; what is open is whether that dynamics *forces* the coefficient $\hbar/2m$ (the factor of 2). DTF therefore *reduces* the Schrödinger equation to one posited constant plus one open coefficient; it does not derive it. *On single-valuedness, two open items, honestly sized*: (i) that the Row-0→Row-1 emergence through the u -clock is a conditioning-type map (which would preserve single-valuedness), still to be shown for the specific u -clock emergence; and (ii) the nodal / multiply-connected case, genuinely open, because there the single-valuedness posit is a real restriction (spinors, anyons). These are smaller than “Wallstrom is unaddressed” — the obstruction is reduced to one declared posit — but larger than a single technical residual.
- **Settled, negatively.** The Bell question of Chain 4 — whether the shared-now joint object is distinguishable from a *fundamental* configuration space — is not owed: it has been answered, and it is not. The guiding wave is on configuration space, and this volume claims no locality beyond the single quantum.

On gravity-mediated entanglement (BMV): *if* the clock-rate constraint is quantised — operator-valued, as QED's non-radiative Coulomb sector is — the guiding-not-driving reading predicts a *positive* outcome. This is stated conditionally, and honestly: the zero-DOF and polarisation results (Volume I–II) treat u as a *classical* constraint, whereas BMV needs it operator-valued — a quantisation this trilogy does not construct. So DTF's BMV stance is “positive, conditional on quantising the constraint,” not a delivered prediction; the framework's delivered falsifiable stakes are in Volume II (GW polarisations, black-hole cores).

The reading offered here is not a separate theory of measurement but the *organizing principle* of Volume I holding in the quantum sector: “becoming definite” is the Row-0→Row-1 step — the *same* clock-definiteness surface whose opposite crossing is the black-hole core of Volume II, where a clock instead de-emerges. Measurement and the singularity are one transition, run in the two directions; that is the sense in which the three volumes are one principle, not three results. A fourth — deep cosmology (the proper now as cosmic time) — is in preparation.

Scripts (public, SymPy/NumPy): DTF_two_clocks_resolved, DTF_guidance_is_spaceface, DTF_phase_amplitude_onecon, DTF_doubleslit_compton_ticks, DTF_born_relaxation_2d, DTF_born_relaxation_2d_mp, DTF_wallstrom_from_row0, DTF_sharednow_counting, DTF_tripartite_ledger, DTF_bell_from_joint_relaxation, DTF_clockjitter_scale, DTF_nelson_from_flexing, DTF_qm_posit_variational, DTF_flexing_time_symmetry.

Provenance of the checks (ontology-first). This volume derives little that is new — that is the point. Its contribution is the *reading*, not new physics, and the scripts are labelled accordingly. **Borrowed** (standard constructions, credited): the relaxation to $|\psi|^2$ (Valentini’s *H*-theorem on Bohm’s guidance) and its joint-configuration extension to $2\sqrt{2}$; the stochastic-quantisation and Nelson checks. **Consistency checks** (a standard fact or definition confirmed, *not* a native derivation): the two-clock *c*-number/state split, the one-constant Planck/Compton identity, the double-slit Compton-tick = de Broglie identity, the singlet purity and Horodecki CHSH [13], the GHZ no-signalling. Two **DTF-native structural** checks earn their place: $u \equiv 1 \Rightarrow \nabla S \neq 0$ (DTF_guidance_is_spaceface), which *refutes* the reading of pilot-wave guidance as a wake in the clock rate, fixing guidance as the space face of the interval; and the Wallstrom reduction (DTF_wallstrom_from_row0) — that single-valuedness reduces to one declared Row-0 posit (a single-valued amplitude assignment) rather than a by-hand quantisation rule, for the simply-connected sector. (The circulation Wallstrom asks about is the *spatial* configuration phase on a fixed slice, not the unbounded temporal one, so no clock-angle argument bears on it; that framing is retired.) The bulk of the native computational content still lives in Volume II; here the work is ontological. None of it rests on circular reasoning.

Appendix A — The DTF Core in brief

For readers coming to this volume first. The Distorted Time Framework takes the local *rate* of time as the one physical primitive, and builds outward in three layers (Volume I).

- **Row 0 — the timeless substrate.** The deepest layer has no clock: the Wheeler–DeWitt constraint $\hat{H}\Psi = 0$ holds with no external time. Correlations are simultaneous, not ordered.
- **Row 1 — the clock.** A definite local clock-rate field $u(\mathbf{x}) > 0$ is conditioned on Row 0; its reading $\tau = \int u dt$ accumulates monotonically — time, and its arrow. A quantum’s phase is this reading, $\Phi = -\omega_C \tau$.
- **Row 2 — space.** An extent is a period radiated at c ($\lambda_C = c T_C$): space is time’s radiation, compiled by an orientation (frame) field. Because c is one universal exchange rate, Lorentz invariance is its signature.
- **One object, four faces.** Extremising the interval gives one phase read four ways: inertia ($\mathbf{p} = \nabla S$), free fall (geodesic), the pilot wave ($\mathbf{v} = \nabla S/m$), and the path integral. Inertia (the rate ω_C) and the pilot wave (the space-face gradient ∇S) are tied by $\mathbf{p} = \nabla S = m\mathbf{v}$, both $\hbar$ -set and linear in m ; the G -coupled well the rate digs is gravity, a separate object.

- **The proper now.** The clock rate does not propagate ($\pi_u \equiv 0$), so it is fixed across space on one preferred slice — a real but empirically inert present, shared by gravity and by entanglement.
- **Definiteness.** “Becoming definite” is the Row-0→1 step; a single clock-definiteness threshold governs measurement (this volume) and the black-hole core (Volume II).

Code and data availability

The SymPy/NumPy verification scripts named throughout (the `DTF_born_relaxation` and `DTF_wallstrom_from_row0` scripts, `DTF_nelson_from_flexing`, `DTF_qm_posit_variational`, `DTF_flexing_time_symmetry`, and the entanglement/counting scripts) are publicly available at <https://github.com/billd-dtf/Distorted-Time-Framework> (release v1.0, matching this revision); see the repository’s README for a script-to-claim index and run instructions.

Author’s Note: the role of AI

The Distorted Time Framework—its central hypothesis (time’s local *rate*, the ADM lapse, as the one physical primitive), its ontology-first strategy, and the physical interpretations—was conceived and developed by the author over several years of independent research outside academic institutions. The mathematical formalization was carried out in collaboration with Claude (Anthropic): the Born-rule relaxation and Wallstrom-reduction arguments, the entanglement, Bell, and functional-content counting results, and the symbolic/numerical verification scripts (SymPy/NumPy) referenced throughout. Separately, other large language models (ChatGPT, Grok, Gemini) were used as independent referees, providing adversarial review of the manuscript; their critiques shaped the revisions but not the derivations.

In each case the author specified the physical target, the constraints the result had to satisfy, and the direction of the argument, and reviewed and approved each result; the AI executed and checked the formal derivations. Without that assistance the author could not have produced the mathematical formalization at this level of detail.

This is disclosed because intellectual honesty requires it. It does not change the epistemic status of the results—a derivation is correct or it is not, independent of how it was produced—but it does mean they warrant careful independent scrutiny rather than assumed authority. The author welcomes that scrutiny; the explicit claim-type labels and the reproducible verification scripts are provided precisely to enable it.

References

- [1] M. Born, “Zur Quantenmechanik der Stoßvorgänge,” *Z. Phys.* **37**, 863–867 (1926).
- [2] D. Bohm, “A Suggested Interpretation of the Quantum Theory in Terms of ‘Hidden’ Variables. I & II,” *Phys. Rev.* **85**, 166–179, 180–193 (1952).
- [3] E. Nelson, “Derivation of the Schrödinger Equation from Newtonian Mechanics,” *Phys. Rev.* **150**, 1079–1085 (1966).
- [4] E. Schrödinger, “Über die kräftefreie Bewegung in der relativistischen Quantenmechanik” (Zitterbewegung), *Sitzungsber. Preuss. Akad. Wiss. Phys.-Math. Kl.* **24**, 418–428 (1930).
- [5] D. Hestenes, “The Zitterbewegung Interpretation of Quantum Mechanics,” *Found. Phys.* **20**, 1213–1232 (1990).
- [6] F. Guerra and L. M. Morato, “Quantization of dynamical systems and stochastic control theory,” *Phys. Rev. D* **27**, 1774–1786 (1983).
- [7] A. O. Caldeira and A. J. Leggett, “Path integral approach to quantum Brownian motion,” *Physica A* **121**, 587–616 (1983).

- [8] E. Madelung, “Quantentheorie in hydrodynamischer Form,” *Z. Phys.* **40**, 322–326 (1927).
- [9] J. S. Bell, “On the Einstein Podolsky Rosen Paradox,” *Physics* **1**, 195–200 (1964).
- [10] T. C. Wallstrom, “Inequivalence between the Schrödinger equation and the Madelung hydrodynamic equations,” *Phys. Rev. A* **49**, 1613–1617 (1994).
- [11] A. Valentini, “Signal-locality, uncertainty, and the subquantum H -theorem. I,” *Phys. Lett. A* **156**, 5–11 (1991).
- [12] B. S. Cirel’son (Tsirelson), “Quantum generalizations of Bell’s inequality,” *Lett. Math. Phys.* **4**, 93–100 (1980).
- [13] R. Horodecki, P. Horodecki, and M. Horodecki, “Violating Bell inequality by mixed spin- $\frac{1}{2}$ states,” *Phys. Lett. A* **200**, 340–344 (1995).
- [14] A. M. Gleason, “Measures on the Closed Subspaces of a Hilbert Space,” *J. Math. Mech.* **6**, 885–893 (1957).
- [15] P. Busch, “Quantum States and Generalized Observables: A Simple Proof of Gleason’s Theorem,” *Phys. Rev. Lett.* **91**, 120403 (2003).
- [16] T. Maudlin, *Quantum Non-Locality and Relativity*, 3rd ed. (Wiley-Blackwell, 2011).
- [17] T. Norsen, *Foundations of Quantum Mechanics* (Springer, 2017).
- [18] D. Dürr, S. Goldstein, T. Norsen, W. Struyve, and N. Zanghì, “Can Bohmian mechanics be made relativistic?” *Proc. R. Soc. A* **470**, 20130699 (2014).
- [19] R. Tumulka, “A Relativistic Version of the Ghirardi–Rimini–Weber Model,” *J. Stat. Phys.* **125**, 821–840 (2006).